Lab # 01

Week 1 Task with C++	
Contents Covered:	C++ Introduction Installation of IDE

What is C++?

- C++ is a cross-platform language that can be used to create high-performance applications.
- C++ was developed by Bjarne Stroustrup, as an extension to the C language.
- C++ gives programmers a high level of control over system resources and memory.
- The language was updated 4 major times in 2011, 2014, 2017, and 2020 to C++11, C++14, C++17, C++20.

Why Use C++?

- C++ is one of the world's most popular programming languages.
- C++ can be found in today's operating systems, Graphical User Interfaces, and embedded systems.
- C++ is an object-oriented programming language which gives a clear structure to programs and allows code to be reused, lowering development costs.
- C++ is portable and can be used to develop applications that can be adapted to multiple platforms.
- C++ is fun and easy to learn!
- As C++ is close to C, C# and Java, it makes it easy for programmers to switch to C++ or vice versa.

Difference between C and C++:

- C++ was developed as an extension of C, and both languages have almost the same syntax.
- The main difference between C and C++ is that C++ support classes and objects, while C does not.

C++ Get Started:

To start using C++, we need two things:

- A text editor, like Notepad, to write C++ code
- A compiler, like GCC, to translate the C++ code into a language that the computer will understand

C++ Install IDE:

An IDE (Integrated Development Environment) is used to edit AND compile the code.

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Popular IDE's include Code::Blocks, Eclipse, and Visual Studio. These are all free, and they can be used to both edit and debug C++ code.

Note: ***Web-based IDE's can work as well, but functionality is limited.***

We will use **Code::Blocks** in our course lab, which we believe is a good place to start.

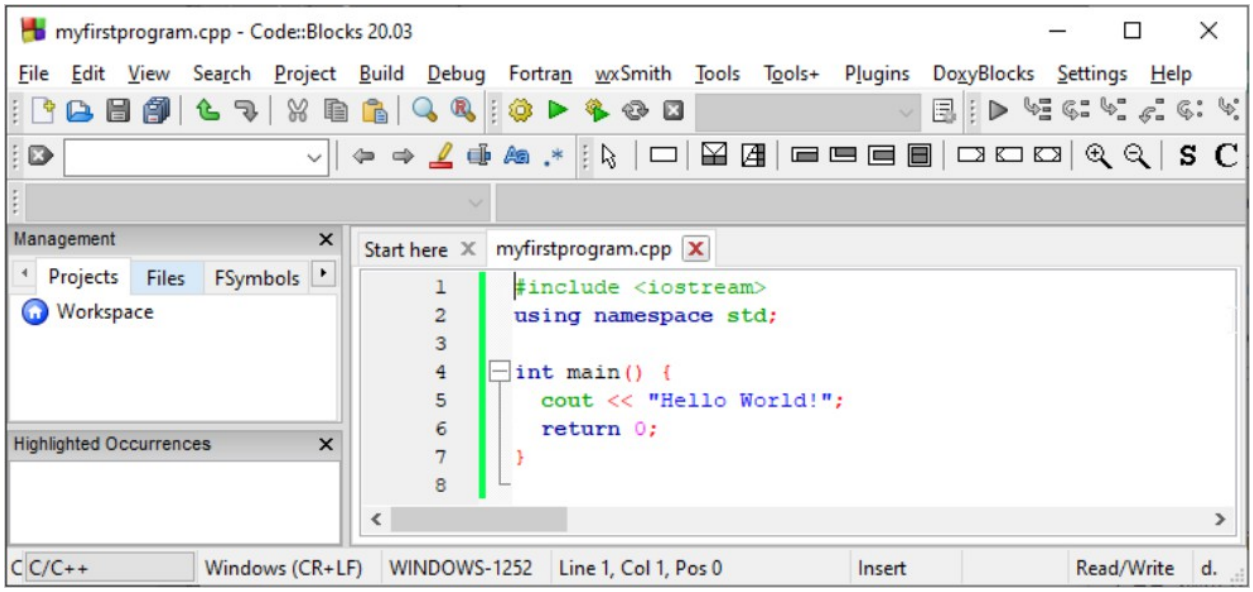
Let's create our first C++ file.

1. Open Codeblocks and go to **File > New > Empty File**.

2. Write the following C++ code and save the file as myfirstprogram.cpp (**File > Save File as**):

Don't worry if you don't understand the code above - we will discuss it in detail in later chapters. For now, focus on how to run the code.

In Codeblocks, it should look like this:



Then, go to **Build > Build and Run** to run (execute) the program. The result will look something to this:

```
Hello World!
Process returned 0 (0x0) execution time : 0.011 s
Press any key to continue.
```


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C++ Syntax:

Let's break up the following code to understand it better:

Example Explained:

Line 1: #include <iostream> is a **header file library** that lets us work with input and output objects, such as cout (used in line 5). Header files add functionality to C++ programs.

Line 2: using namespace std means that we can use names for objects and variables from the standard library.

Line 3: A blank line. C++ ignores white space. But we use it to make the code more readable.

Line 4: Another thing that always appear in a C++ program is int main(). This is called a **function**. Any code inside its curly brackets {} will be executed.

Line 5: cout (pronounced "see-out") is an **object** used together with the *insertion operator* (<<) to output/print text. In our example, it will output "Hello World!".

Note: Every C++ statement ends with a semicolon ;.

Note: The body of int main() could also been written as:

```
int main () {

cout << "Hello World! ";

return 0;

}
```

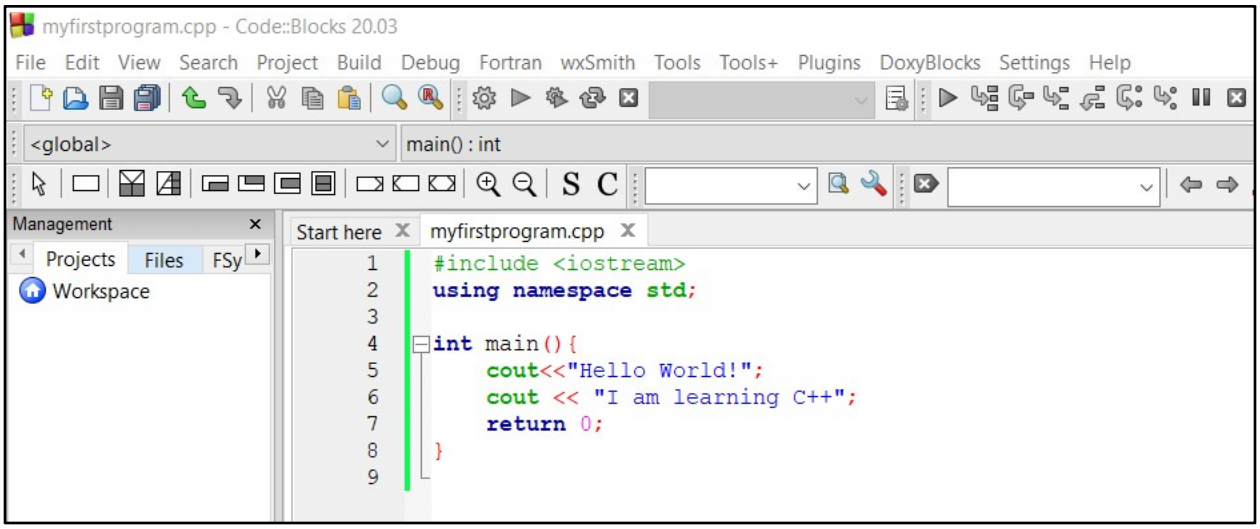
Line 6: return 0 ends the main function.

Line 7: Do not forget to add the closing curly bracket } to actually end the main function.

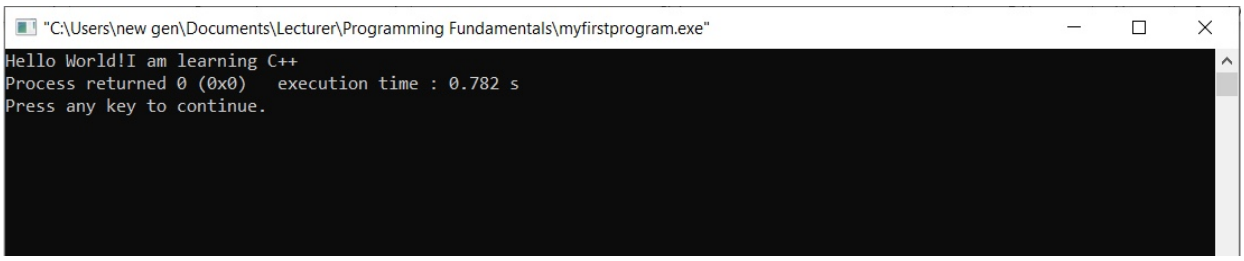
You can add as many cout objects as you want. However, note that it does not insert a new line at the end of the output:

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Example 1:



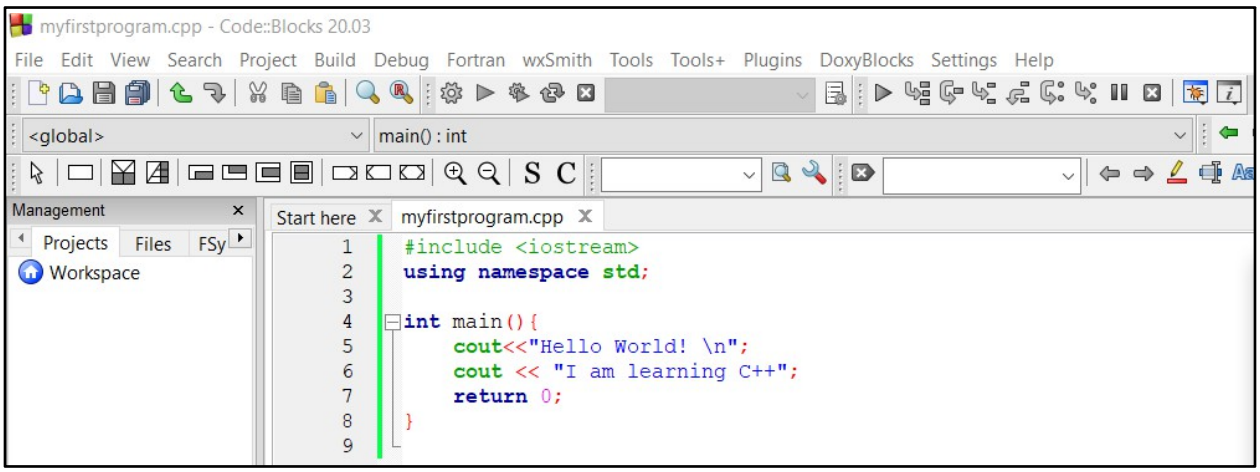
Output:



New Lines:

To insert a new line, you can use the \n character:

Example 2:



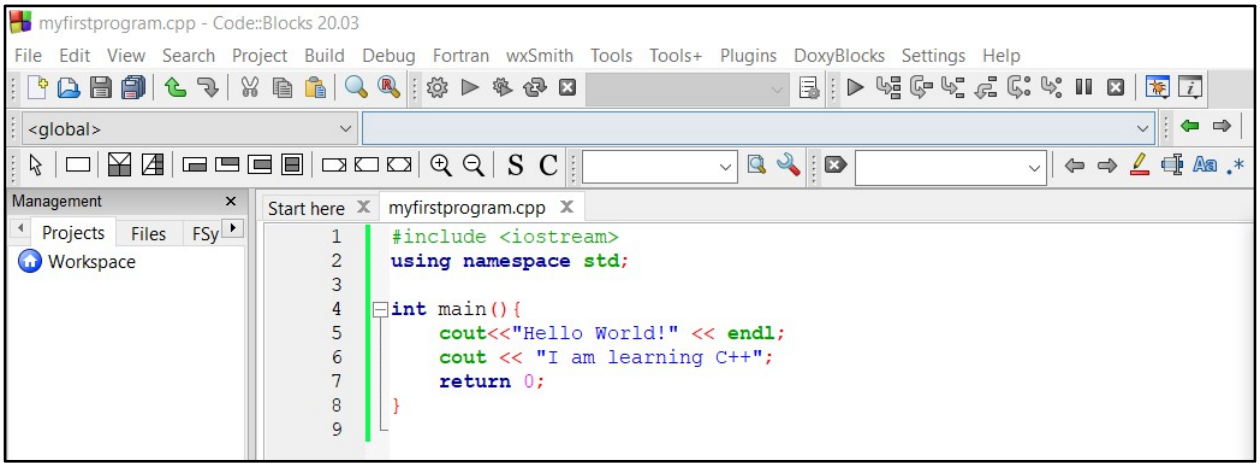
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Output:

```
"C:\Users\new gen\Documents\Lecturer\Programming Fundamentals\myfirstprogram.exe"
Hello World!
I am learning C++
Process returned 0 (0x0)   execution time : 0.077 s
Press any key to continue.
```

Another way to insert a new line, is with the endl manipulator:

Example 3:



Output:

```
"C:\Users\new gen\Documents\Lecturer\Programming Fundamentals\myfirstprogram.exe"
Hello World!
I am learning C++
Process returned 0 (0x0)   execution time : 0.088 s
Press any key to continue.
```

Both \n and endl are used to break lines. However, \n is most used.

But what is \n exactly?

The newline character (\n) is called an **escape sequence**, and it forces the cursor to change its position to the beginning of the next line on the screen. This results in a new line.

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Examples of other valid escape sequences are:

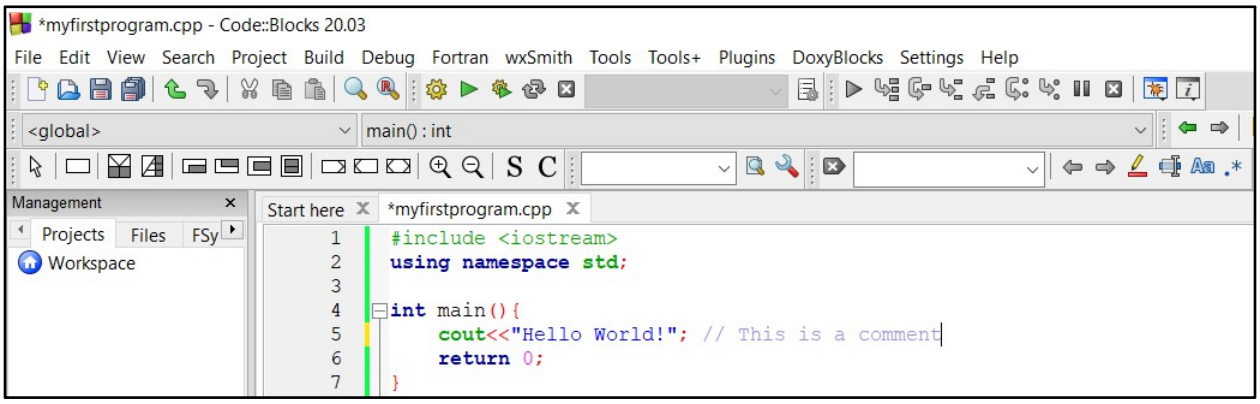
Escape Sequence	Description
\t	Creates a horizontal tab
\\	Inserts a backslash character (\)
\"	Inserts a double quote character

C++ Comments:

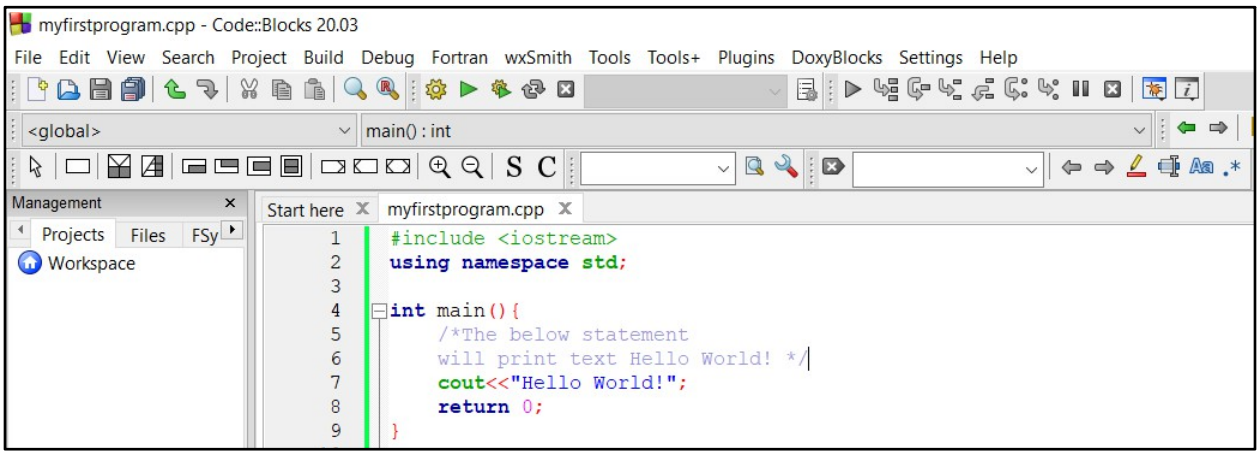
Comments can be used to explain C++ code, and to make it more readable. It can also be used to prevent execution when testing alternative code. Comments can be singled-lined or multi-lined.

Single-line comments start with two forward slashes (//).

Any text between // and the end of the line is ignored by the compiler (will not be executed).



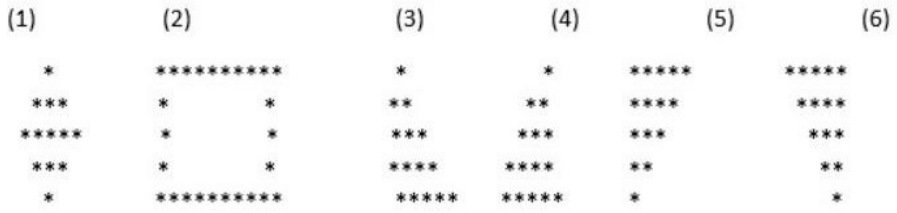
Multi-line comments start with /* and ends with */.



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Test Exercise

1. Print the following shape using simple print statements.



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